

定义:

扩散通量定义: $J\phi D = -\mu \nabla \phi$

Ω_c 第 c 个控制体

$V_c = |\Omega_c|$ 第 c 个控制体的体积

F_c 属于控制 c 的全部面组成的集合

$\vec{n}_{cf}$ 面 f 相对于控制体 c 的单位外法向量

$\vec{S}_{cf} = |S_f| \vec{n}_{cf}$ 面 f 相对于控制体 c 的局部有向面积向量

O_f, N_f 内部面 f 的 owner 单元与 neighbour 单元.

$\vec{n}_f$ 从 O_f 指向 N_f 的全局单位法向量.

$\vec{S}_f = |S_f| \vec{n}_f$ 从 O_f 指向 N_f 的全局有向面积向量

σ_{cf} 面 f 相对于控制体 c 的关联符号: $\sigma_{cf} = \begin{cases} 1 & c = O_f \\ -1 & c = N_f \end{cases}$

$\vec{d}_f = x_{Nf} - x_{Of}$ 从 owner 中心指向 neighbour 中心的向量

$d_f = |\vec{d}_f|$ 从 owner 中心到 neighbour 中心的距离

ϕ_c ϕ 在控制体 c 的平均值 $\phi_c = \frac{1}{V_c} \int_{V_c} \phi(x) dv$

D_f 面 f 的离散扩散传递系数

H_f 按 $O_f \rightarrow N_f$ 定向的全局数值扩散通量

$H_{cf} = \sigma_{cf} H_f$ 内部面 f 相对于控制体 c 的局部向外物理扩散通量

H_{cb} 边界面 b 相对于所属控制体的向外物理扩散通量

R_c 控制体 c 上 $\nabla \cdot (\mu \nabla \phi)$ 的体积

$$01 \quad \frac{\partial \phi}{\partial t} + \nabla \cdot (u\phi) = 0$$

有限体积的半离散格式:

$$\Omega = \bigcup_{c=1}^{N_c} \Omega_c \quad \text{其中 } \Omega_c \cap \Omega_d = \emptyset \quad (\forall c \neq d)$$

$$\text{定义 } \phi_c(t) = \frac{1}{V_c} \int_{\Omega_c} \phi(x, t) dV \quad \text{则有 } \int_{\Omega_c} \phi(x, t) dV = V_c \phi_c(t)$$

$$\text{对 } \frac{\partial \phi}{\partial t} + \nabla \cdot (u\phi) = 0 \quad \text{在 } \Omega_c \text{ 上积分得 } \int_{\Omega_c} \frac{\partial \phi}{\partial t} dV + \int_{\Omega_c} \nabla \cdot (u\phi) dV = 0$$

$$\text{其中 } \int_{\Omega_c} \frac{\partial \phi}{\partial t} dV = \frac{d}{dt} \int_{\Omega_c} \phi dV = \frac{d}{dt} (V_c \phi_c) = V_c \frac{d\phi_c}{dt}$$

$$\text{其中 } \int_{\Omega_c} \nabla \cdot (u\phi) dV \quad \underline{\text{高斯定理}} \quad \oint_{\partial \Omega_c} (u\phi) \cdot \vec{n}_{cf} dS = \sum_{f \in F_c} \int_{S_f} (u\phi) \cdot \vec{n}_{cf} dS$$

$$\text{代入得 } V_c \frac{d\phi_c}{dt} + \sum_{f \in F_c} \int_{S_f} (u\phi) \cdot \vec{n}_{cf} dS = 0$$

$$V_c \frac{d\phi_c}{dt} + \sum_{f \in F_c} (u_f \cdot \vec{s}_{cf}) \phi_f = 0 \quad \text{记 } F_{cf} = u_f \cdot \vec{s}_{cf}$$

$$V_c \frac{d\phi_c}{dt} + \sum_{f \in F_c} F_{cf} \phi_f = 0$$

$$\frac{d\phi_c}{dt} = - \frac{1}{V_c} \sum_{f \in F_c} F_{cf} \phi_f$$

显示空间离散:

$$\text{- 迎风: } \phi_{cf}^{up} = \begin{cases} \phi_{of} & F_{cf} \geq 0 \\ \phi_{nf} & F_{cf} < 0 \end{cases}$$

$$\text{记 } F_{cf}^+ = \max\{F_{cf}, 0\} \quad F_{cf}^- = \min\{F_{cf}, 0\}$$

$$F_{cf} \cdot \phi_{cf}^{up} = F_{cf}^+ \phi_{of} + F_{cf}^- \phi_{nf}$$

数值耗散 (泰勒展开)

$$\text{故有 } \frac{d\phi_c}{dt} = - \frac{1}{V_c} \sum_{f \in F_c} (F_{cf}^+ \phi_{of} + F_{cf}^- \phi_{nf})$$

$$\text{记 } R_c(\phi) = \frac{\sum_{f \in F_c} (F_{cf}^+ \phi_{of} + F_{cf}^- \phi_{nf})}{V_c}$$

$$\text{则 } \frac{d\phi_c}{dt} = -R_c(\phi)$$

显示时间离散:

$$\phi_c(t^n + \Delta t) = \phi_c(t^n) + \Delta t \frac{d\phi_c}{dt} \Big|_{t^n} + o(\Delta t)$$

$$-R_c(\phi_n) = \frac{d\phi_c}{dt} \Big|_{t^n} \approx \frac{\phi_c^{n+1} - \phi_c^n}{\Delta t}$$

$$\text{其中 } \phi_c^{n+1} = \phi_c^n - \Delta t R_c(\phi_n)$$

$$\Delta t \leq \frac{C_{max}}{\max_{f \in F_c} \frac{(F_{cf})}{V_c}}$$

02 diffusion equation

$$\frac{\partial \phi}{\partial t} - \nabla \cdot (\mu \nabla \phi) = 0$$

半离散形式:

$$\text{令 } \Omega = \bigcup_{c=1}^{N_c} \Omega_c \quad \text{其中 } \Omega_c \cap \Omega_d = \emptyset \quad (\forall c \neq d) \quad \text{记 } \phi_c(t) = \frac{1}{V_c} \int_{\Omega_c} \phi(x, t) dV$$

$$\text{在任意控制体 } \Omega_c \text{ 上积分有 } \int_{\Omega_c} \frac{\partial \phi}{\partial t} dV - \int_{\Omega_c} \nabla \cdot (\mu \nabla \phi) dV = 0 \quad (1)$$

$$\text{其中 } \int_{\Omega_c} \frac{\partial \phi}{\partial t} dV = \frac{d}{dt} \int_{\Omega_c} \phi dV = \frac{d}{dt} (V_c \phi_c) = V_c \frac{d\phi_c}{dt} \quad (2)$$

$$\text{其中 } \int_{\Omega_c} \nabla \cdot (\mu \nabla \phi) dV \xrightarrow{\text{高斯散度定理}} \oint_{\partial \Omega_c} \mu \nabla \phi \cdot \vec{n} ds \xrightarrow{\text{可加性}} \sum_{f \in F_c} \int_{S_f} \mu \nabla \phi \cdot \vec{n}_{cf} ds$$

$$\xrightarrow{\text{面心一点积分}} \sum_{f \in F_c} \mu_f (\nabla \phi)_f \cdot \vec{S}_{cf} \quad (3)$$

$$\text{故将 (2) (3) 代入 (1) 得原式为 } V_c \frac{d\phi_c}{dt} - \sum_{f \in F_c} \mu_f (\nabla \phi)_f \cdot \vec{S}_{cf} = 0 \quad (4)$$

$$\text{由定义 } H_f = \nabla \phi \cdot \vec{S}_f \cdot \vec{S}_f = -\mu_f (\nabla \phi)_f \cdot \vec{S}_f \quad (5)$$

$$\text{代入得 } V_c \frac{d\phi}{dt} - \sum_{f \in F_c} H_f = 0 \quad V_c \frac{d\phi}{dt} + \left(\sum_{f \in F_{c \text{ int}}} \nabla_{cf} H_f + \sum_{b \in F_{c \text{ bound}}} H_{cb} \right) = 0$$

有

$$\frac{d\phi}{dt} = - \frac{1}{V_c} \left(\sum_{f \in F_{c \text{ int}}} \nabla_{cf} H_f + \sum_{b \in F_{c \text{ bound}}} H_{cb} \right) \triangleq R_c(\phi) \quad (6)$$

$$\text{故原式等价于 } \frac{d\phi}{dt} = R_c(\phi)$$

$$\text{另: } R_c(\phi) = - \frac{1}{V_c} \sum_{f \in F_c} H_f = - \frac{1}{V_c} \sum_{f \in F_c} \mu_f (\nabla \phi)_f \cdot \vec{S}_f \xrightarrow{\text{倒回}} \frac{1}{V_c} \int_{\Omega_c} \nabla \cdot (\mu \nabla \phi) dV$$

定义 $[\nabla \cdot (\mu \nabla \phi)]_c$ (Open Foam 中)

显示空间离散: $H_f = -\mu_f (\nabla \phi)_f \cdot \vec{S}_f$

(一) 对于内部面 $\vec{S}_f = \vec{E}_f + \vec{T}_f$ 其中 $\vec{E}_f \parallel d\vec{f}$ 即正交网格 记 $\vec{e}_f = \frac{d\vec{f}}{df}$

$$(\nabla \phi)_f \cdot \vec{S}_f = (\nabla \phi)_f \cdot \vec{E}_f + (\nabla \phi)_f \cdot \vec{T}_f$$

$$\text{其中正交部分 } (\nabla \phi)_f \cdot \vec{E}_f = |\vec{E}_f| (\nabla \phi)_f \cdot \vec{e}_f \approx \frac{|\vec{E}_f|}{df} (\phi_{nf} - \phi_{of}) \quad \text{面法向梯度用差分近似}$$

其中反对称部分 $(\nabla \phi)_f \cdot \vec{T}_f$

$$\text{其中 } (\nabla \phi)_f \text{ 由梯度插值 } (\nabla \phi)_f = g_{of} \boxed{\nabla \phi_{of}} + g_{nf} \nabla \phi_{nf} \text{ 确定}$$

其中 $g_{of} + g_{nf} = 1$

$$\boxed{\nabla \phi_c} \xrightarrow{\text{定义}} \frac{1}{V_c} \int_{\Omega_c} \nabla \phi_c dV \xrightarrow{\text{高斯梯度定理}} \frac{1}{V_c} \oint_{\partial \Omega} \phi \cdot \vec{n} ds$$

$$\approx \frac{1}{V_c} \sum_{f \in F_c} \boxed{\phi_f} \cdot \vec{S}_f$$

内部面值采用线性插值 $\boxed{\phi_f} = g_{of} \phi_{of} + g_{nf} \phi_{nf}$ 其中 $g_{of} + g_{nf} = 1$

故若记 $D_f = \mu_f \frac{|\vec{e}_f|}{\Delta f}$ $C_f = \mu_f (\nabla \phi)_f \cdot \vec{T}_f$ 则有内部面数值扩散通量为

$$H_f = D_f (\phi_{of} - \phi_{nf}) - C_f$$

(二) 对边界面

Dirichlet 边界条件

若边界面 b 给定 $\phi_b = \phi_{\text{specified}}(x_b, t)$, 则有

$$(\nabla \phi)_b \cdot \vec{n}_{cb} \approx \frac{\phi_b - \phi_c}{d_{cb}}$$

$$H_{cb} = -\mu_b (\nabla \phi)_b \cdot \vec{s}_{cb} = \frac{\mu_b |s_b|}{d_{cb}} (\phi_c - \phi_b)$$

Neumann 边界条件

若给定外法向导数 $\frac{\partial \phi}{\partial n} = g_b$

$$H_{cb} = -\mu_b g_b |s_b|$$

显示时间离散:

在 t^n 处对 ϕ_c 作向前 Taylor 展开:

$$\phi_c(t^n + \Delta t) = \phi_c(t^n) + \Delta t \left. \frac{d\phi_c}{dt} \right|_{t^n} + \frac{\Delta t^2}{2} \left. \frac{d^2\phi_c}{dt^2} \right|_{t^n} + \dots$$

$$\left. \frac{d\phi_c}{dt} \right|_{t^n} = \frac{\phi_c^{n+1} - \phi_c^n}{\Delta t} + o(\Delta t) \approx \frac{\phi_c^{n+1} - \phi_c^n}{\Delta t} = R_c(\phi^n)$$

$$\text{故 } \phi_c^{n+1} = \phi_c^n + \Delta t R_c(\phi^n)$$

其本为使所有控制体稳定, 全局时间步取 $\Delta t \leq \min_c \frac{V_c}{\sum_{f \in F_c} D_f}$

实际引入安全系数 $0 < \sigma < 1$ $\Delta t = \sigma \min_c \frac{V_c}{\sum_{f \in F_c} D_f}$

03 对流扩散方程

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (u \phi) - \nabla \cdot (\mu \nabla \phi) = 0$$

有限体积法半离散形式:

将 $\frac{\partial \phi}{\partial t} + \nabla \cdot (u \phi) - \nabla \cdot (\mu \nabla \phi) = 0$ 在 Ω_c 上积分得

$$\int_{\Omega_c} \frac{\partial \phi}{\partial t} dv + \int_{\Omega_c} \nabla \cdot (u \phi) dv - \int_{\Omega_c} \nabla \cdot (\mu \nabla \phi) dv = 0$$

其中 $\int_{\Omega_c} \frac{\partial \phi}{\partial t} dv = \frac{d}{dt} \int_{\Omega_c} \phi dv = \frac{d}{dt} (V_c \phi_c) = V_c \frac{d\phi_c}{dt}$

其中 $\int_{\Omega_c} \nabla \cdot (u \phi) dv \xrightarrow{\text{高斯散度定理}} \oint_{\partial \Omega_c} (u \phi) \cdot \vec{d\vec{b}} \xrightarrow{\text{可加}} \sum_{f \in F_c} \oint_{S_f} (u \phi) \cdot \vec{n}_f ds$
 $\xrightarrow{\text{高斯积分近似}} \sum_{f \in F_c} (u_f \phi_f) \cdot \vec{S}_f = \sum_{f \in F_c} (u_f \cdot \vec{S}_f) \phi_f$

记 $F_{cf} \triangleq u_f \cdot \vec{S}_f$ 原式 = $\sum_{f \in F_c} F_{cf} \phi_f$

其中 $\int_{\Omega_c} \nabla \cdot (\mu \nabla \phi) dv \xrightarrow{\text{高斯散度定理}} \oint_{\partial \Omega_c} \mu \nabla \phi \cdot \vec{d\vec{b}} = \sum_{f \in F_c} \oint_{S_f} \mu \nabla \phi \cdot \vec{n}_f ds$
 $= \sum_{f \in F_c} \mu_f (\nabla \phi)_f \cdot \vec{S}_f$

故原方程的有限体积半离散骨架为

$$V_c \frac{d\phi_c}{dt} + \sum_{f \in F_c} F_{cf} \phi_f - \sum_{f \in F_c} \mu_f (\nabla \phi)_f \cdot \vec{S}_f = 0$$

$$\frac{d\phi_c}{dt} = -\frac{1}{V_c} \left[\sum_{f \in F_c} F_{cf} \phi_f - \sum_{f \in F_c} \mu_f (\nabla \phi)_f \cdot \vec{S}_f \right]$$

$$= -\frac{1}{V_c} \left[\sum_{f \in F_{c, \text{int}}} \nabla \phi_f (F_{cf} \phi_f - \mu_f (\nabla \phi)_f \cdot \vec{S}_f) + \sum_{b \in F_{c, \text{bound}}} (F_{cb} \phi_b - \mu_b (\nabla \phi)_b \cdot \vec{S}_b) \right]$$

$$\triangleq R_c(\phi) = [-\nabla \cdot (u \phi) + \nabla \cdot (\mu \nabla \phi)]_c$$

空间显式离散:

(一) 面值 ϕ_f 采用一阶迎风插值

$$\phi_{cf}^{up} = \begin{cases} \phi_{of} & F_f \geq 0 \\ \phi_{nf} & F_f < 0 \end{cases}$$

(二) 对于内部面

扩散项 $H_f = -\mu_f (\nabla \phi)_f \cdot \vec{S}_f$ $\vec{S}_f = \vec{E}_f + \vec{T}_f$ 其中 $\vec{E}_f \parallel \vec{d}_f$ 记 $\vec{e}_f = \frac{\vec{d}_f}{|\vec{d}_f|}$

其中正交部分 面法向标度采用中心差分近似

其交叉扩散项 $(\nabla \phi)_f \cdot \vec{T}_f$

其 $(\nabla \phi)_f$ 由梯度插值 $(\nabla \phi)_f = g_{of} \nabla \phi_{of} + g_{nf} \nabla \phi_{nf}$

$$\nabla \phi_c = \frac{1}{V_c} \int_{\Omega_c} \nabla \phi_c dV = \frac{1}{V_c} \oint_{\partial c} \phi \cdot \vec{n} dS \approx \frac{1}{V_c} \sum_{f \in F_c} \phi_f \cdot \vec{S}_{cf}$$

ϕ_f 采用线性插值

$$\phi_f = g_{of} \phi_{of} + g_{nf} \phi_{nf} \quad \text{其中 } g_{of} + g_{nf} = 1$$

若记 $D_f = \mu_f \frac{|\vec{S}_f|}{\Delta f}$ $C_f = \mu_f (\nabla \phi)_f \cdot \vec{T}_f$ 则有内部面数值扩散量为

$$H_f = D_f (\phi_{of} - \phi_{nf}) - C_f$$

(三) 对于边界面

Dirichlet 边界条件

若边界面 b 给定 $\phi_b = \phi_{\text{specified}}(x_b, t)$ 则

$$H_{cb} = -\mu_b (\nabla \phi)_b \cdot \vec{S}_{cb} = \frac{\mu_b |S_b|}{d_{cb}} (\phi_c - \phi_b)$$

Neumann 边界条件

若给定外法向导数 $\frac{\partial \phi}{\partial n} = g_b$

$$H_{cb} = -\mu_b g_b |S_b|$$

显式时间离散 (同时受 CFL 和 扩散 Fourier 限制)

$$\phi_c(t^n + \Delta t) = \phi_c(t^n) + \Delta t \frac{d\phi_c}{dt} \Big|_{t^n} + o(\Delta t)$$

$$\phi_c^{n+1} \approx \phi_c^n + \Delta t R_c(\phi_n)$$

$$\Delta t = \nabla \min_c \frac{V_c}{\sum_{f \in F_c} [\max(F_{cf}, 0) + D_{cf}]} \quad 0 < \nabla < 1$$

(1) 显式时间推进 + 对流通量计算

(2) 显式时间推进 + 扩散面通量

(3) 显式时间推进 + 对流残差和扩散残差组合

(4) 离散算子自动构造方程组, 再求解未知场。 稳定,

04 Poisson 方程

$$\nabla \cdot \nabla \phi = w \quad \text{即} \quad \nabla^2 \phi = w$$

$$\underline{\text{FVM}} :: \text{laplacian}(\phi) = w$$

有限体积半离散

$$\text{在控制体 } \Omega_c \text{ 上积分: } \int_{\Omega_c} \nabla \cdot \nabla \phi \, dV = \int_{\Omega_c} w \, dV$$

$$\text{其中 } \int_{\Omega_c} \nabla \cdot \nabla \phi \, dV \quad \underline{\text{Gauss 散度定理}} \quad \oint_{\partial \Omega_c} \nabla \phi \cdot \vec{dS} \quad \underline{\text{对面, 用 Gauss 积分}} \quad \sum_{f \in F_c} (\nabla \phi)_f \cdot \vec{S}_{cf}$$

$$\text{由 } w_c = \frac{1}{V_c} \int_{\Omega_c} w \, dV, \text{ 代入得} \quad \sum_{f \in F_c} (\nabla \phi)_f \cdot \vec{S}_{cf} = V_c w_c$$

$$\sum_{f \in F_{c, \text{int}}} \nabla \phi_f \cdot \vec{S}_{cf} + \sum_{b \in F_{c, \text{bound}}} (\nabla \phi)_b \cdot \vec{S}_{cb} = V_c w_c$$

空间离散: 将面梯度 $(\nabla \phi)_f$ 由单元值封闭, 从而转成代数方程组, 从而求数值解.

(一) 内部面 f:

$$\vec{S}_f = \vec{E}_f + \vec{T}_f \quad \text{其中 } \vec{E}_f \parallel \vec{d}_f$$

对于正交部分 $\vec{E}_f$:

$$(\nabla \phi)_f \cdot \vec{E}_f \propto \frac{|\vec{E}_f|}{d_f} (\phi_{nf} - \phi_{of})$$

对于反对称部分: $\vec{T}_f \quad (\nabla \phi)_f \cdot \vec{T}_f$

$$\text{其中 } (\nabla \phi)_f \text{ 由本梯度插值} \quad (\nabla \phi)_f = g_{of} \nabla \phi_{of} + g_{nf} \nabla \phi_{nf}$$

$$\nabla \phi_c = \frac{1}{V_c} \int_{\Omega_c} \nabla \phi_c \, dV = \frac{1}{V_c} \oint_{\partial \Omega_c} \phi \cdot \vec{n} \, dS \propto \frac{1}{V_c} \sum_{f \in F_c} \phi_f \cdot \vec{S}_{cf}$$

ϕ_f 采用线性插值

$$\phi_f = g_{of} \phi_{of} + g_{nf} \phi_{nf} \quad \text{其中 } g_{of} + g_{nf} = 1$$

$$\text{则: } (\nabla \phi)_f \cdot \vec{S}_{cf} = \frac{|\vec{E}_f|}{d_f} (\phi_{nf} - \phi_{of}) + (\nabla \phi)_f \cdot \vec{T}_f$$

(二) 边界面 b.

Dirichlet 边界面通量

$$\phi_b = \phi_{\text{specified}}(\chi_b)$$

$$\text{则 } (\nabla \phi)_b \cdot \vec{S}_{cb} = \frac{|\vec{E}_{cb}|}{d_{cb}} (\phi_b - \phi_c) + (\nabla \phi)_b \cdot \vec{T}_{cb}$$

05 Navier-Stokes 方程

$$\nabla \cdot \mathbf{u} = 0$$

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u} \otimes \mathbf{u}) = -\frac{1}{\rho} \nabla p + \nu \nabla \cdot (\mu \nabla \mathbf{u})$$

$\mathbf{u}$ 速度矢量 $\mathbf{u}^n$ 上-时间步速度 $\mathbf{u}^*$ 预测中间速度 $\mathbf{u}^{n+1}$ 更新后最终速度

μ 动力黏度 ρ 密度 p 物理压力 $\nu = \frac{\mu}{\rho}$ 运动黏度

$\pi = \frac{p}{\rho}$ 运动学压力 H_f^{adv}, H_f^{diff} 全局定向的对流, 黏性动量通量

于是原方程变为 $\nabla \cdot \mathbf{u} = 0$

投影法:

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u} \otimes \mathbf{u}) = -\nabla \pi + \nabla \cdot (\nu \nabla \mathbf{u})$$

注: 压力没有独立的演化方程, 而是作为 Lagrange 乘子, 使新速度落入无散空间 $V_0 = \{\mathbf{v} : \nabla \cdot \mathbf{v} = 0\}$

设预测速度为 $\mathbf{u}^* = \mathbf{u}^n + \Delta t \nabla \pi^n$ 于是 $\mathbf{u}^{n+1} = \mathbf{u}^* - \Delta t \nabla \pi^{n+1}$ (预测速度, 减掉压力梯度带来的那一部分, 剩下无散真实速度 $\nabla \cdot \mathbf{u}^{n+1} = 0$)

$$\xrightarrow{\text{同时求散度}} \nabla \cdot \mathbf{u}^{n+1} = \nabla \cdot \mathbf{u}^* - \Delta t \nabla \cdot (\nabla \pi^{n+1}) \xrightarrow{\nabla \cdot \mathbf{u}^{n+1} = 0} 0 = \nabla \cdot \mathbf{u}^* - \Delta t \nabla^2 \pi^{n+1}$$

$$\Rightarrow \nabla^2 \pi^{n+1} = \frac{1}{\Delta t} \nabla \cdot \mathbf{u}^* \quad (\text{泊松方程}) \quad [\text{解动量预测方程 } \mathbf{u}^* \rightarrow \pi^{n+1} \rightarrow \mathbf{u}^{n+1}] \star$$

注: $\frac{\partial \mathbf{u}}{\partial t} \approx \frac{\mathbf{u}^* - \mathbf{u}^n}{\Delta t}$ 丢 $-\nabla \pi$ (在下步更新) ($\nabla \cdot \mathbf{u}^* \neq 0$)

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u} \otimes \mathbf{u}) = -\nabla \pi + \nabla \cdot (\nu \nabla \mathbf{u})$$

$$\Rightarrow \frac{\mathbf{u}^* - \mathbf{u}^n}{\Delta t} + \nabla \cdot (\mathbf{u}^n \otimes \mathbf{u}^*) - \nabla \cdot (\nu \nabla \mathbf{u}^*) = 0 \quad [\text{预测方程}]$$

$$\Rightarrow \nabla^2 \pi^{n+1} = \frac{1}{\Delta t} \nabla \cdot \mathbf{u}^* \Rightarrow \mathbf{u}^{n+1} = \mathbf{u}^* - \Delta t \nabla \pi^{n+1} \quad (\Delta t \text{ 自己设置步长})$$

注: 当 Dirichlet 边界条件时 假设 $\mathbf{u}^{n+1} = \mathbf{g}^{n+1}$

$$\mathbf{u}^{n+1} = \mathbf{u}^* - \Delta t \nabla \pi^{n+1}$$

$$\mathbf{u}^{n+1}|_{\text{边界}} = \mathbf{g}^{n+1} \quad \text{则} \quad \mathbf{g}^{n+1} = \mathbf{u}^* - \Delta t \nabla \pi^{n+1} \Rightarrow \mathbf{g}^{n+1} \cdot \vec{n} = \mathbf{u}^* \cdot \vec{n} - \Delta t \nabla \pi^{n+1} \cdot \vec{n}$$

$$\Rightarrow \frac{\Delta \pi^{n+1}}{\Delta t} = \frac{1}{\Delta t} (\mathbf{u}^* - \mathbf{g}^{n+1}) \cdot \vec{n}$$

有限体积半离散开式:

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u} \otimes \mathbf{u}) - \nabla \cdot (\nu \nabla \mathbf{u}) + \nabla \pi = 0$$

$$\int_{\Omega_c} \frac{\partial \mathbf{u}}{\partial t} dV + \int_{\Omega_c} \nabla \cdot (\mathbf{u} \otimes \mathbf{u}) dV - \int_{\Omega_c} \nabla \cdot (\nu \nabla \mathbf{u}) dV + \int_{\Omega_c} \nabla \pi dV = 0$$

$$\text{其中} \int_{\Omega_c} \frac{\partial \mathbf{u}}{\partial t} dV \approx V_c \frac{d\mathbf{u}_c}{dt}$$

$$\int_{\Omega_c} \nabla \cdot (\mathbf{u} \otimes \mathbf{u}) dV = \oint_{\partial \Omega_c} (\mathbf{u} \otimes \mathbf{u}) \cdot \vec{n} ds \approx \sum_{f \in F_c} (\mathbf{u}_f \cdot \mathbf{S}_c f) \vec{n}_f$$

$$\int_{\Omega_c} \nabla \cdot (\sqrt{\nabla u}) dV \approx \sum_{f \in F_c} \sqrt{f} (\nabla u)_f \cdot \vec{S}_f$$

$$\int_{\Omega_c} \nabla \pi dV \approx V_c (\nabla \pi)_c$$

记 $H_f^{adv} = F_f u_f^{adv}$ $H_f^{diff} = \sqrt{f} (\nabla u)_f \cdot \vec{S}_f$ 代入得

$$V_c \frac{d u_c}{dt} + \sum_{f \in F_c}^{int} (\sigma_{cf} H_f^{adv} - \sigma_{cf} H_f^{diff}) + \sum_{b \in F_c}^{bnd} (H_b^{adv} - H_b^{diff}) + V_c (\nabla \pi)_c = 0$$

(H_f^{adv} H_f^{diff} $(\nabla \pi)_c$ 需由后面给出)。

关于插值等:

1) 对于 $H_f^{adv} = F_f u_f^{adv}$ 而言
 u_f 采用-迎风: $u_f^up = \begin{cases} u_{of} & F_f \geq 0 \\ u_{nf} & F_f < 0 \end{cases}$
 记 $(F)^+ = \max(F, 0)$ $(F)^- = \min(F, 0)$

$$H_f^{adv} = F_f^+ u_{of} + F_f^- u_{nf}$$

2) 对于 $H_f^{diff} = \sqrt{f} (\nabla u)_f \cdot \vec{S}_f$ 而言 (和前面一样)

$$\vec{S}_f = \vec{E}_f + \vec{T}_f$$

$$H_f^{diff} \approx \sqrt{f} \frac{|\vec{E}_f|}{df} (u_{nf} - u_{of}) + \sqrt{f} (\nabla u)_f \cdot \vec{T}_f \rightarrow \text{Gauss linear corrected}$$

(三) $(\nabla \pi)_c \approx \frac{1}{V_c} \sum_{f \in F_c} \pi_f S_{cf}$ $\pi_f \approx \lambda_f \pi_{of} + (1 - \lambda_f) \pi_{nf}$

$$(\nabla \pi)_f \cdot \vec{S}_f \approx \frac{|\vec{E}_f|}{df} (\pi_{nf} - \pi_{of}) + (\nabla \pi)_f \cdot \vec{T}_f$$

此时①是关于 p, u^n 的方程

先把压力顶丢了, 解方程: $u_c^* \rightarrow F_f^* \rightarrow \pi_{n+1} \rightarrow F_f^{n+1} \rightarrow u_{c,n+1}^*$

时间离散与投影压力-速度耦合

先把①中新压力顶去掉 (逻辑和前注中相同, 表达式本质上一样)

$$\frac{V_c}{\Delta t} (u_c^* - u_c^n) + \sum_{f \in F_c}^{int} (\sigma_{cf} H_f^{adv*} - \sigma_{cf} H_f^{diff*}) + \sum_{b \in F_c}^{bnd} (H_b^{adv*} - H_b^{diff*}) = 0 \quad (\text{解 } u_c^*)$$

其中 $H_f^{adv*} = F_f^n u_f^*$ $F_f^n = u_f^n \cdot \vec{S}_f$ 但是 $\nabla \cdot u^* \neq 0$ 即 $\sum_{f \in F_c} \sigma_{cf} F_f^* \neq 0$

通过 u_c^* 用一致面插值-迎风 $F_f^* = u_f^* \cdot \vec{S}_f$ 得 F_f^* (匹配设置)

代入 $\sum_{f \in F_c} (\nabla \pi^{n+1})_f \cdot \vec{S}_{cf} = \frac{1}{\Delta t} \sum_{f \in F_c} F_{cf}^*$ (泊松方程) 得 $\nabla \pi^{n+1}$

修正面通量 $F_f^{n+1} = F_f^* - \Delta t (\nabla \pi^{n+1})_f \cdot \vec{S}_f$

u_c^* 代入 $\nabla^2 \pi^{n+1} = \frac{1}{\Delta t} \nabla \cdot u^*$ 得 π^{n+1}
 先不用, 极限下一样

$$u^{n+1} = u^* - \Delta t \nabla \pi^{n+1}$$

PIOS法:

有限体积半离散形式:

$$\frac{\partial u}{\partial t} + \nabla \cdot (u \otimes u) - \nabla \cdot (\nu \nabla u) + \nabla \pi = 0$$

$$\int_{\Omega_c} \frac{\partial u}{\partial t} dV + \int_{\Omega_c} \nabla \cdot (u \otimes u) dV - \int_{\Omega_c} \nabla \cdot (\nu \nabla u) dV + \int_{\Omega_c} \nabla \pi dV = 0$$

$$\# \int_{\Omega_c} \frac{\partial u}{\partial t} dV \approx V_c \frac{du_c}{dt}$$

$$\int_{\Omega_c} \nabla \cdot (u \otimes u) dV = \oint_{\partial \Omega_c} (u \otimes u) \cdot \vec{n} ds \approx \sum_{f \in F_c} (u_f \cdot S_{cf}) u_f$$

$$\int_{\Omega_c} \nabla \cdot (\nu \nabla u) dV \approx \sum_{f \in F_c} \nu_f (\nabla u)_f \cdot \vec{S}_{cf}$$

$$\int_{\Omega_c} \nabla \pi dV \approx V_c (\nabla \pi)_c$$

$$\text{记 } H_f^{\text{adv}} = F_f u_f^{\text{adv}} \quad H_f^{\text{diff}} = \nu_f (\nabla u)_f \cdot \vec{S}_f \quad \text{代入得}$$

$$V_c \frac{du_c}{dt} + \sum_{f \in F_c}^{\text{int}} (\sigma_{cf} H_f^{\text{adv}} - \sigma_{cf} H_f^{\text{diff}}) + \sum_{b \in F_c}^{\text{bnd}} (H_b^{\text{adv}} - H_b^{\text{diff}}) + V_c (\nabla \pi)_c = 0$$

(H_f^{adv} H_f^{diff} $(\nabla \pi)_c$ 需由后面给出)。

空间离散, 面插值:

1) 对于 $H_f^{\text{adv}} = F_f u_f^{\text{adv}}$ 而言

$$u_f \text{ 采用-迎风插值: } u_f^{\text{up}} = \begin{cases} u_{of} & F_f \geq 0 \\ u_{nf} & F_f < 0 \end{cases}$$

$$\text{记 } (F)^+ = \max(F, 0) \quad (F)^- = \min(F, 0)$$

$$H_f^{\text{adv}} = F_f^+ u_{of} + F_f^- u_{nf}$$

2) 对于 $H_f^{\text{diff}} = \nu_f (\nabla u)_f \cdot \vec{S}_f$ 而言

$$\vec{S}_f = \vec{E}_f + \vec{T}_f$$

$$H_f^{\text{diff}} \approx \nu_f \frac{|\vec{E}_f|}{df} (u_{nf} - u_{of}) + \nu_f (\nabla u)_f \cdot \vec{T}_f$$

$$(3) (\nabla \pi)_c \approx \frac{1}{V_c} \sum_{f \in F_c} \pi_f S_{cf} \quad \pi_f \approx \lambda_f \pi_{of} + (1 - \lambda_f) \pi_{nf}$$

$$(\nabla \pi)_f \cdot \vec{S}_f \approx \frac{|\vec{E}_f|}{df} (\pi_{nf} - \pi_{of}) + (\nabla \pi)_f \cdot \vec{T}_f$$

(同上) 压力校正后应直接修正 F_f , 不能只修正 u_c .

在 $t^n \rightarrow t^{n+1}$ 采用 Euler 时间离散, 并以当前最新通量 f^b 线性化对流项

$$\frac{V_c}{\Delta t} (u_c^{n+1} - u_c^n) + \sum_{f \in F_c} F_{cf}^b u_f^n - \sum_{f \in F_c} V_f (\nabla u^n)_f \cdot \vec{S}_{cf} + \underbrace{V_c (\nabla \pi^b)_c}_{\rightarrow \text{保留}} = 0$$

已知 $V_c, \Delta t, u_c^n, F_{cf}^b, V_f, S_{cf}, (\nabla \pi^b)_c$ 未知 $u_c^{n+1}, u_f^{n+1}, (\nabla u^{n+1})_f$

由插值得到, 方式同上。

将本单元未知 u_c^{n+1} 的全部系数合并, 记 a_c^u ; 把各个邻居单元未知 $u_{\text{other}(c,f)}^{n+1}$ 的系数记 a_{cf}^u

插值之后全部已知项打包 b_c^u

$$\text{故原式等于 } a_c^u u_c^{n+1} + \sum_{f \in F_c} a_{cf}^u u_{\text{other}(c,f)}^{n+1} = b_c^u - V_c (\nabla \pi^b)_c \quad (\text{单行}) \rightarrow \text{拼起来动量矩阵}$$

$$A u^{n+1} = b - V \nabla \pi^b$$

$$\text{记 } H_c^u = b_c^u - \sum_{f \in F_c} a_{cf}^u u_{\text{other}(c,f)}^{n+1} \quad \text{记 } h_c = \frac{H_c^u}{a_c^u} \quad \text{且 } \psi_c = \frac{V_c}{a_c^u}$$

$$\text{得 } u_c^{n+1} = h_c - \psi_c (\nabla \pi)_c$$

$$u_f^* = h_{cf} - \psi_{cf} (\nabla \pi)_f$$

用 π^b 解 $A u^* = b - V \nabla \pi^b$ 得 u^* ;

$$u = u^* - \psi \nabla \pi' \quad \text{代入 } \sum_f u \cdot S_f = 0 \quad \text{得 } \sum_f (\psi_f \nabla \pi'_f) \cdot S_f = \sum_f u_f^* \cdot S_f \quad \pi'$$

$$\text{压力修正 } \pi^{\text{new}} = \pi^b + \pi' \quad u_c' = \frac{H_c[u^* + u'] - H_c[u^*]}{a_c^u} - \psi_c (\nabla \pi')_c \quad \text{且 } \psi_c$$

$$\text{得 } u_c^{\text{new}} = u_c^* - \psi_c (\nabla \pi')_c$$

$$A + \pi^b \rightarrow u^* \rightarrow u_f^* \rightarrow \pi' \rightarrow \pi^{\text{new}} = \pi^b + \pi' \quad u^{\text{new}} = u^* - \psi_c \nabla \pi'$$